Bryan Hu

Contact Information

University of California, San Diego
Department of Mathematics
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Research Interests

Algebraic number theory, representation theory, automorphic forms.

Education

University of California, San Diego

Ph.D. Candidate in Mathematics, June 2026

University of California, Los Angeles

B.S. in Mathematics of Computation, 2016–2020
M.A. in Mathematics, 2018–2020

Papers

Math:

1. Hu, "Exceptional Maass-Shimura Operators for $\mathrm{SU}(2,1)$ in G_2", *preprint* (pdf)
2. Getz, Gutiérrez Terradillos, Hosseinijafari, Hu, Lee, Slipper, Tomé, Yao, Zhao, "Modulation Groups", *submitted* (arxiv)
3. Ariaz, Creech, Hu, Khunger, Kozioł, Rankothge, Zhang, "Iwasawa Algebras of pro-p Iwahori Subgroups", *accepted, Journal de Théorie des Nombres de Bordeaux, 2026* (arxiv)
4. Hu, Johnson-Leung, McGlade, Pollack, Roy, "Algberaicity of Eisenstein Series on Exceptional Groups and Deligne's Conjecture for G_2", *in preparation*

Machine Learning:

1. Yang, Bhat, **Hu**, Cao, Dehmamy, Walters, Yu, "Discovering Symbolic Differential Equations with Symmetry Invariants", *Transactions on Machine Learning Research (TMLR), 2026* (paper)

Conference and Seminar Talks

2026 April	TORA XV (invited talk)
2025 October	University of Arizona Algebra and Number Theory Seminar (invited talk)
2025 October	Souther California Number Theory Day (lightning talk)
2025 August	Aarhus Automorphic Forms Conference (lightning talk)
2025 May	UCSD Number Theory Seminar
2024 May	UCSD Number Theory Seminar

Mentorship and Outreach

2024 Fall	UCSD AWM Mentorship Coordinator
2024 Summer	UCSD AGANT Directed reading for undergrads (Algebraic Number Theory)
2024 Spring	UCSD AGANT Directed reading for undergrads (Class Field Theory)
2023 Summer	Mentor, Math Summer Program for High School Students
2023 Spring	UCSD AGANT Directed reading for undergrads (Elliptic Curves)

	2022 Spring	UCSD AGANT Directed reading for undergrads (Quaternion Algebras)
	2021–2022	UCSD Math Graduate Student Council Mentorship Committee
	2021 Summer	Teaching Assistant, Awesome Math Summer Program (Number Theory 2 and Abstract Algebra)
	2021 Spring	UCSD AGANT Directed reading for undergrads (Elliptic Curves)
	2021 Spring	Teaching Assistant, Euler Circle (Complex Analysis)
	2020 Fall	Teaching Assistant, Euler Circle (Markov Chains)
	2017–2019	Assistant Instructor, Los Angeles Math Circle
Teaching Experience	2026 Winter	TA, Statistical Methods (Math 183)
	2025 Fall	TA, Calculus III (Math 10C)
	2025 Spring	TA, Applied Complex Analysis (Math 120B)
	2025 Winter	TA, Discrete Mathematics and Graph Theory (Math 154)
	2024 Fall	Instructor, Linear Algebra (Math 18)
	2024 Spring	Tutor Training TA
	2024 Winter	Tutor Training TA
	2023 Fall	Lead TA, Calculus I (Math 10A)
	2023 Spring	TA, Intro to Stochastic Processes (Math 180B)
	2023 Winter	TA, Abstract Algebra II (Math 100B)
	2022 Fall	TA, Abstract Algebra I (Math 100A)
	2022 Spring	TA, Abstract Algebra III (Math 100C)
	2022 Winter	TA, Enumerative Combinatorics (Math 184)
	2021 Fall	TA, Calculus for Science and Engineering (Math 20A)
	2021 Summer	TA, Linear Algebra (Math 18)
	2021 Summer	TA, Vector Calculus (Math 20E)
	2021 Spring	TA, Enumerative Combinatorics (Math 184)
	2021 Winter	TA, Calculus II (Math 10B)
	2020 Fall	TA, Differential Equations (Math 20D)
Honors and Awards	2020-2025	James B. Ax Fellowship (UCSD)
	2020	Sherwood Prize (UCLA), For overall outstanding achievement and exceptional performance in mathematics at the undergraduate level
	2020	Departmental Scholar (UCLA), Most advanced program offered to mathematics majors; joint B.S. and M.A. degree completed in 4 years
	2020	Phi Beta Kappa
	2016–2020	Regents Scholarship (UCLA)
Conferences and Workshops	2026 Sept	AIM SQuaREs Workshop (Quaternionic Modular Forms)
	2025 August	Aarhus Automorphic Forms
	2025 March	Arizona Winter School (Representation Theory of p-adic Groups)
	2024 Sept	AIM SQuaREs Workshop (Quaternionic Modular Forms)
	2024 July	Rethinking Number Theory 5 (p-adic Iwasawa algebras)
	2023 May	CIRM Research School and Conference (Arithmetic Statistics)
	2022 July	APAW Graduate Instructional Workshop

2022 March	Arizona Winter School (Automorphic Forms Beyond GL_2)
2021 July	PCMI Graduate Summer School (Number Theory Informed by Computation)